



RIDDHIMAN MOULICK

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riddhiman-m.github.io

EDUCATION

Indian Institute of Technology, Kharagpur <i>CGPA: 9.04/10.00</i>	2020 – Present
Major: Electrical Engineering	
Minor: Computer Science & Engineering	
Micro: Artificial Intelligence and Applications	
City International School, Pune <i>Grade: 96.8%</i>	2018 – 2020
Higher Secondary (CBSE)	
The Bishop's School, Undri, Pune <i>Grade: 98.4%</i>	2006 – 2018
Secondary School (ICSE)	

WORK EXPERIENCE

Convolutional Prompting for Continual Learning IIT Kharagpur	Aug' 2023 – Dec' 2023
Bachelor Thesis Project <i>Supervisor: Prof. Abir Das (CSE, IIT Kharagpur)</i>	
- Devised a novel convolution-based, <i>on-the-fly prompting</i> technique to overcome the challenge of <i>catastrophic forgetting</i> in the class-incremental continual learning setup in vision models. - Experimented with various task-similarity metrics based on <i>class-label</i>, <i>image</i> and <i>class-attribute embeddings</i> and chose the class attributes' similarity metric for a parameter-efficient prompt pool expansion. - Achieved a performance improvement of $\sim 3\%$ over SOTA approaches on benchmark datasets while also <i>reducing the parameter overhead</i> by $\sim 50\%$ (Paper accepted at CVPR 2024 Conference).	
Becky Recommendations Oracle	May 2023 – Jul' 2023
Cloud Applications Developer Intern	
- Added the user action recommendation feature in <i>Becky</i>, a digital assistant, to automatically initiate a contextual conversation with users, and suggest actions based on their possible requirements. - Fetched various users' pods and analyzed the data obtained from these, to understand the prospective needs of the user and to alert the user about delayed pods or action updates via mail.	

PROJECTS AND RESEARCH

Computing Game-Theoretic Equilibria <i>Course Project: Algorithmic Game Theory</i>	Fall 2022
- Acquired all the parameterized inputs in normal form games by parsing the input from Gambit GUI - Computed and enlisted the various strategies for Dominant strategy equilibria and Nash equilibria	
Navigating a Frozen Lake with Stochastic Wind <i>Course Project: Reinforcement Learning</i>	Fall 2023
- Applied Policy Evaluation, Iteration, and Monte-Carlo Algorithms for agent navigation - Visualized the learning iterations and the frozen lake environment using the OpenAI <i>gym</i> package	

COURSEWORK

Computer Architecture and Organisation, Algorithms-I, Deep Learning Foundations and Applications, Reinforcement Learning, Algorithmic Game Theory, AI for Economics, Information Security, Database Management Systems

SKILLS

- Programming Languages:** Python, C/C++, Java, SQL, Verilog, Assembly, $\LaTeX$
- Tools & Libraries:** PyTorch, Tensorflow, Keras, OpenCV, NumPy, Matplotlib, Pandas, Matlab, Simulink, Stateflow

AWARDS AND ACHIEVEMENTS

- Selected to receive the **Chanakya Graduate Fellowship, 2024** by AI4ICPS, IIT Kharagpur
- Achieved an **All India Rank 6** and city rank 1 in ICSE Class X with an aggregate score of **98.4%**
- Awarded the **Silver Certificate** for securing **AIR 233** in Technothon-2016, conducted by IIT Guwahati

[*CV last updated on April 2, 2024]